

Knowledge Engineering for Concordant Systems / Knowledge Graphs for AI

CIS 590 - 890, Kansas State University

Spring 2026

Course Description:

Effective approaches for integration and analysis of multi-disciplinary data require knowledge-based semantic models that are suitable as a framework for integration, while avoiding overly extreme simplification of domain knowledge. A major challenge in developing such semantic models is the knowledge acquisition and representation bottleneck.

In this class, we will focus on using systematic methodologies for knowledge acquisition and knowledge engineering for generating semantic models as well as knowledge graphs that work together, in harmony, concordantly. Our examples will come from biomedical, food and agriculture domains when exploring real-world examples of how some of the most diverse datasets are handled with the current state of the art. We will explore how the frameworks we generate can also be used in different applications, including systems that use artificial intelligence, for in-depth analysis of multi-disciplinary, diverse datasets and data sources.

Instructor:

Dr. Hande Küçük McGinty

hande@ksu.edu

Office: DUE 2167

Office hours: Tuesday- Thursday, 10 – 11:30 am CST or by appointment

Class Hours:

Tue-Th 3:55 PM- 5:20 PM DUE1116

Reporting on project progress will be done as part of meetings between the instructor and the students. The meetings will take place during class hours.

Join Zoom Meeting

<https://ksu.zoom.us/j/3597002666?pwd=eEQvaUVoZ25oWFZHRjlBSnY5TGhtQT09>

Meeting ID: 359 700 2666

Passcode: 131747

Prerequisites:

CIS 560 Database System Concepts

Course Materials:

Background reading material will be provided via Canvas or email.

Method of Instruction:

The class will be thought as lectures as described in the outline. On the marked weeks, Students will review papers and present reporting on their class projects.

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Evaluation:

Quizzes: 10%
Paper Reviews: 10%
Project Proposal: 20%
Project Midterm review report: 20%
Final Project report: 20%
Project code:20%

Course Outline:

Week 1-2
Introduction to Semantic Web

Week 3
Project Proposal Discussions / Introduction to Building Concordant Systems using KNowledge Acquisition and Representation Methodology (KNARM)

Week 4,
Introduction to Building Concordant Systems using KNowledge Acquisition and Representation Methodology (KNARM) / Meet with domain experts get started on projects

Week 6
Project Proposal Presentations / Paper assignments

Week 7:
NSF ProtoOKN presentations

Week 8: :
Review of Methods for Semantic Web Applications

Week 9: Spring break

Week 10:
Review of Methods for Semantic Web Applications

Week 11
Paper review and Project Progress Presentations

Week 12-13
Evolving Concordant Systems for AI and other use cases

Week 14-15
Evolving Concordant Systems for AI and other use cases

Week 16
Final Project Presentations and Discussion

Week 17:
Make up classes and Final Project Reports Due

Known Remote Class Schedule:

Feb 3rd and Feb 5th
(Maybe) Feb 19th
May 5th and 7th

Important Dates:**March 15-22 : Spring break**

May 8th – Last day of classes

May 14th – Final project reports are due

May 20th – Final Grades are due

Statement Regarding Academic Honesty

Kansas State University has an Honor and Integrity System based on personal integrity, which is presumed to be sufficient assurance that, in academic matters, one's work is performed honestly and without unauthorized assistance. Undergraduate and graduate students, by registration, acknowledge the jurisdiction of the Honor and Integrity System. The policies and procedures of the [Honor and Integrity System](#) apply to all full and part-time students enrolled in undergraduate and graduate courses on-campus, off-campus, and via distance learning. A component vital to the Honor and Integrity System is the inclusion of the Honor Pledge which applies to all assignments, examinations, or other course work undertaken by students. The Honor Pledge is implied, whether or not it is stated: "On my honor, as a student, I have neither given nor received unauthorized aid on this academic work." A grade of XF can result from a breach of academic honesty. The F indicates failure in the course; the X indicates the reason is an Honor Pledge violation.

Statement Regarding Students with Disabilities

At K-State it is important that every student has access to course content and the means to demonstrate course mastery. Students with disabilities may benefit from services including accommodations provided by the Student Access Center. Disabilities can include physical, learning, executive functions, and mental health. You may register at the [Student Access Center](#) or to learn more contact:

Manhattan/Olathe/K-State Online – Student Access Center

accesscenter@k-state.edu

785-532-6441

K-State Salina Campus – Julie Rowe; Student Success Coordinator

jarowe@k-state.edu

785-820-7908

Students already registered with the Student Access Center please request your Letters of Accommodation early in the semester to provide adequate time to arrange your approved academic accommodations. Once SAC approves your Letter of Accommodation it will be e-mailed to you, and your instructor(s) for this course. Please follow up with your instructor to discuss how best to implement the approved accommodations.

Statement Defining Expectations for Classroom Conduct

All student activities in the University, including this course, are governed by the [Student Judicial Conduct Code](#) as outlined in the Student Governing Association By Laws, Article V, Section 3, number 2. Students who engage in behavior that disrupts the learning environment may be asked to leave the class.

Statement on Mutual Respect and Inclusion in K-State Teaching and Learning Spaces

At K-State, faculty and staff are committed to creating and maintaining an inclusive and supportive learning environment for students from diverse backgrounds and perspectives. K-State courses, labs, and

other virtual and physical learning spaces promote equitable opportunity to learn, participate, contribute, and succeed, regardless of age, race, color, ethnicity, nationality, genetic information, ancestry, disability, socioeconomic status, military or veteran status, immigration status, Indigenous identity, gender identity, gender expression, sexuality, religion, culture, as well as other social identities.

Faculty and staff are committed to promoting equity and believe the success of an inclusive learning environment relies on the participation, support, and understanding of all students. Students are encouraged to share their views and lived experiences as they relate to the course or their course experience, while recognizing they are doing so in a learning environment in which all are expected to engage with respect to honor the rights, safety, and dignity of others in keeping with the K-State [Principles of Community](#).

If you feel uncomfortable because of comments or behavior encountered in this class, you may bring it to the attention of your instructor, advisors, and/or mentors. If you have questions about how to proceed with a confidential process to resolve concerns, please contact the [Student Ombudsperson Office](#). Violations of the [student code of conduct](#) can be reported using the [Code of Conduct Reporting Form](#). You can also report [discrimination, harassment or sexual harassment](#), if needed.

Statement Regarding Discrimination, Harassment, and Sexual Harassment

Kansas State University is committed to maintaining academic, housing, and work environments that are free of discrimination, harassment, and sexual harassment. Instructors support the University's commitment by creating a safe learning environment during this course, free of conduct that would interfere with your academic opportunities. Instructors also have a [duty to report](#) any behavior they become aware of that potentially violates the University's policy prohibiting discrimination, harassment, and sexual harassment, as outlined by [PPM 3010](#).

If a student is subjected to discrimination, harassment, or sexual harassment, they are encouraged to make a non-confidential report to the University's [Office of Civil Rights and Title IX \(OCR & TIX\)](#) using the [online reporting form](#). Incident disclosure is not required to receive resources at K-State. Reports that include domestic and dating violence, sexual assault, or stalking, should be considered for reporting by the complainant to the [Kansas State University Police Department](#) or the [Riley County Police Department](#). Reports made to law enforcement are separate from reports made to OCR & TIX. A complainant can choose to report to one or both entities. Confidential support and advocacy can be found with the [K-State Center for Advocacy, Response, and Education \(CARE\)](#). Confidential mental health services can be found with [Lafene Counseling and Psychological Services \(CAPS\)](#). Academic support can be found with the [Student Support and Accountability \(SSA\)](#). SSA is a non-confidential resource. OCR & TIX also provides a [comprehensive list of resources](#) on their website. If you have questions about non-confidential and confidential resources, please contact OCR & TIX at civilrights@k-state.edu or (785) 532-6220.

Academic Freedom Statement

Kansas State University is a community of students, faculty, and staff who work together to discover new knowledge, create new ideas, and share the results of their scholarly inquiry with the wider public. Although new ideas or research results may be controversial or challenge established views, the health and growth of any society requires frank intellectual exchange. Academic freedom protects this type of free exchange and is thus essential to any university's mission.

Moreover, academic freedom supports collaborative work in the pursuit of truth and the dissemination of knowledge in an environment of inquiry, respectful debate, and professionalism. Academic freedom is not limited to the classroom or to scientific and scholarly research, but extends to the life of the university as

well as to larger social and political questions. It is the right and responsibility of the university community to engage with such issues.

Weapons Policy Statement

Kansas State University prohibits the possession of firearms, explosives, and other weapons on any University campus, with certain limited exceptions, including the lawful concealed carrying of handguns, as provided in the University [Weapons Policy](#).

You are encouraged to take the [online weapons policy education module](#) to ensure you understand the requirements of the policy, including the requirements related to concealed carrying of handguns on campus. Students possessing a concealed handgun on campus must be lawfully eligible to carry and either at least 21 years of age or a licensed individual who is 18-21 years of age. All carrying requirements of the policy must be observed in this class, including but not limited to the requirement that a concealed handgun be completely hidden from view, securely held in a holster that meets the specifications of the policy, carried without a chambered round of ammunition, and that any external safety be in the “on” position.

If an individual carries a concealed handgun in a personal carrier such as a backpack, purse, or handbag, the carrier must remain within the individual’s exclusive and uninterrupted control. This includes wearing the carrier with a strap, carrying or holding the carrier, or setting the carrier next to or within the immediate reach of the individual.

Each individual who lawfully possesses a handgun on campus shall be wholly and solely responsible for carrying, storing and using that handgun in a safe manner and in accordance with the law, Board policy and University policy. All reports of suspected violation of the weapons policy are made to the University Police Department by picking up any Emergency Campus Phone or by calling 785-532-6412.

Campus Safety Statement

Kansas State University is committed to providing a safe teaching and learning environment for student and faculty members. In order to enhance your safety in the unlikely case of a campus emergency make sure that you know where and how to quickly exit your classroom and how to follow any emergency directives. Current Campus Emergency Information is available at the [University’s Advisory](#) webpage.

Student Resources Statement

K-State has many resources to help contribute to student success. These resources include accommodations for academics, paying for college, student life, health and safety, and others. Check out the [Student Guide to Help and Resources: One Stop Shop](#) for more information.

Student Academic Creations

Student academic creations are subject to Kansas State University and Kansas Board of Regents Intellectual Property Policies. For courses in which students will be creating intellectual property, the K-State policy can be found at [University Handbook, Appendix R: Intellectual Property Policy and Institutional Procedures \(part I.E.\)](#). These policies address ownership and use of student academic creations.

8. Statement on Mental Health

Your mental health and good relationships are vital to your overall well-being. Symptoms of mental health issues may include excessive sadness or worry, thoughts of death or self-harm, inability to concentrate, lack of motivation, or substance abuse. Although problems can occur anytime for anyone, you should pay extra attention to your mental health if you are feeling academic or financial stress, discrimination, or have experienced a traumatic event, such as loss of a friend or family member, sexual assault or other physical or emotional abuse.

If you are struggling with these issues, do not wait to seek assistance.

Language for Kansas State Manhattan Campus:

Kansas State University [Counseling and Psychological Services](#) offers free and confidential services to assist you to meet these challenges.

[Lafene Health Center](#) has specialized nurse practitioners to assist with mental health.

The [Division of Academic Success and Student Affairs](#) can direct you to additional resources.

[K-State Family Center](#) offers individual, couple, and family counseling services on a sliding fee scale.

[Center for Advocacy, Response, and Education \(CARE\)](#) provides free and confidential assistance for those in our K-State community who have been victimized by violence.

The [Division of Academic Success and Student Affairs](#) can direct you to additional resources.

University Excused Absences

K-State has a [University Excused Absence policy](#) (Section F62). Class absence(s) will be handled between the instructor and the student unless there are other university offices involved. For university excused absences, instructors shall provide the student the opportunity to make up missed assignments, activities, and/or attendance specific points that contribute to the course grade, unless they decide to excuse those missed assignments from the student's course grade. Please see the policy for a complete list of university excused absences and how to obtain one. Students are encouraged to contact their instructor regarding their absences.

Copyright Statement

Class, syllabus, and lecture notes as well as presentation slides for this course carry a copyright 2026 Dr. Hande McGinty. During this course students are prohibited from selling notes to or being paid for taking notes by any person or commercial firm without the express written permission of the professor teaching this course. In addition, students in this class are not authorized to provide class notes or other class-related materials to any other person or entity, other than sharing them directly with another student taking the class for purposes of studying, without prior written permission from the professor teaching this course. K-State Student Conduct Code prohibits any illegal or unauthorized taking, selling, or distribution of class notes. Students violating this provision will be subject to discipline under the conduct code, including but not limited to possible expulsion from K-State.